

Walter Thomas Moya Araya

Junior Data Scientist | Junior Data Analyst | Quantitative Analysis

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Professional Summary

Industrial Engineer with an MSc in Engineering Sciences. Experience in ETL and data quality at PwC Chile, quantitative development at ClickAlgo, and independent research in financial machine learning.

Technical Skills

Data and BI: Python (Pandas, NumPy), SQL, R, Excel, Power BI, ETL and exploratory data analysis.

Machine learning: scikit-learn, TensorFlow/Keras, clustering, PCA, time series and temporal validation.

Cloud and tools: Google Cloud (BigQuery, Data Fusion, Cloud Storage), C#/.NET, Git, Docker, MLflow.

Professional Experience

Quantitative ML Developer | Independent Research

10/2024 – Present

- Designed a modular financial ML platform in Python with microstructure features, fractional differentiation, CUSUM sampling, triple-barrier labeling and meta-labeling.
- Implemented nested hyperparameter search, walk-forward and CPCV with purging and feature-memory embargo; sample weighting by uniqueness and time decay.
- Integrated backtesting with costs and probability-based position sizing, deflated Sharpe and pre-registered forward testing; artifact provenance and hash-chained signal records.

Junior Programmer | ClickAlgo, United Kingdom · Remote

11/2023 – 10/2024

- Developed quantitative indicators in C#/.NET using clustering, Kalman filters and volatility analysis; delivered tested, documented modules and provided technical support to international users.
- Published a technical article on minimum-correlation portfolios using Pearson, Spearman and Kendall.

Analyst Trainee | PwC Chile

08/2022 – 11/2022

- Optimized historical inventory loads of hundreds of thousands of records to **under 30 seconds per file**.
- Supported ETL pipelines on Google Cloud using Python and REST APIs, schema validation and data quality checks for Power BI dashboards.

Selected Projects

K-Means Indicator v2 | Applied Machine Learning

ClickAlgo · 2025

Developed a market regime indicator using K-means clustering, PCA and price, volatility and momentum features. Refactored the code and added visualizations and Telegram alerts.

FSG Ultimate / Smart Grid Ultimate v2.1 | Automation and Risk

cTrader Store · 2024–2026

Developed and published a mean-reversion trading robot in C#/.NET with closed-bar execution, ATR-based targets and three layers of risk controls to manage exposure and losses.

Smart Portfolio Architect | Portfolio Diversification

ClickAlgo · 2024

Automated minimum-correlation asset selection in C# for cTrader. The tool analyzes historical data and compares asset combinations to support portfolio diversification.

Portfolio Optimizer | Python, Dash and Backtesting

GitHub · 2025

Built a Dash application with four allocation methods and rebalancing backtests. Added return, drawdown and correlation charts for strategy comparisons.

Education

MSc in Engineering Sciences, Industrial Engineering

08/2021 – 06/2023

Universidad Diego Portales · **Highest distinction** · Thesis grade: 6.8/7.0.

Thesis: construction machinery detection, tracking and activity quantification (YOLOv8, BoT-SORT).

Industrial Engineering Degree

03/2016 – 07/2023

Universidad Diego Portales · Graduated with distinction · Grade: 5.7/7.0.

Selected Certifications

MLOps Specialization — Duke University (2026) · Machine Learning — DeepLearning.AI / Stanford (2025).

IBM Data Science Professional Certificate (2025) · Google Data Analytics Professional Certificate (2025).